usv_hydro

Physical Models and Formulas

Technical Reference Report

Project	Maritime Autonomous Systems Simulator
Package	`usv_hydro` v0.1.0
Platform	ROS 2 Jazzy + Gazebo Sim 8
Date	April 2026
Status	Phase 1 Complete – 21/21 tests passed

This report provides a detailed description of each sub-model's input/output definitions, the physical formulas applied, configuration parameters, and inter-model data flow.

Source files:

`HydroTypes.cc`, `HydroCellSystem.cc`, `BuoyancyModel.cc`, `DragModel.cc`, `HydroIntegrator.cc`, `EnvironmentModel.cc`

Contents

1 Introduction and System Architecture

1.1 Overview

usv_hydro is a C++ physics plugin running on Gazebo Sim 8. At each simulation step (ISystemPreUpdate) it reads body kinematics from Gazebo, evaluates six physics sub-models, and injects the resulting forces and moments back into the Gazebo physics engine via AddWorldWrench.

1.2 Inputs from Gazebo

Table 1: Input variables read from Gazebo at each step

Variable	Type	Description
$\mathbf{p}_W$	$\mathbb{R}^3$	Body position in world frame (m)
$\mathbf{q}_W$	quaternion	World-frame orientation
$\mathbf{v}^W$	$\mathbb{R}^3$	Linear velocity in world frame (m/s)
$\boldsymbol{\omega}^W$	$\mathbb{R}^3$	Angular velocity in world frame (rad/s)
Δt	scalar	Simulation step duration (s)
t	scalar	Total simulation time (s)

The plugin derives body-frame velocity and acceleration:

$$\boldsymbol{\nu} = \begin{bmatrix} \mathbf{R}^T \mathbf{v}^W \\ \mathbf{R}^T \boldsymbol{\omega}^W \end{bmatrix} \in \mathbb{R}^6 \tag{1}$$

$$\dot{\boldsymbol{\nu}} \approx \frac{\boldsymbol{\nu}(t) - \boldsymbol{\nu}(t - \Delta t)}{\Delta t} \in \mathbb{R}^6 \tag{2}$$

where $\mathbf{R}$ is the world→body rotation matrix (from $\mathbf{q}_W$).

DOF ordering is consistent across all models:

$$[0\text{=surge},\ 1\text{=sway},\ 2\text{=heave},\ 3\text{=roll},\ 4\text{=pitch},\ 5\text{=yaw}]$$

1.3 Total Force Composition

Forces computed each step are applied via AddWorldWrench:

$$\boxed{\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{buoy}} + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{stiff}} + \mathbf{F}_{\text{rad}} + \mathbf{F}_{\text{exc}}} \tag{3}$$

1.4 Data-Flow Diagram

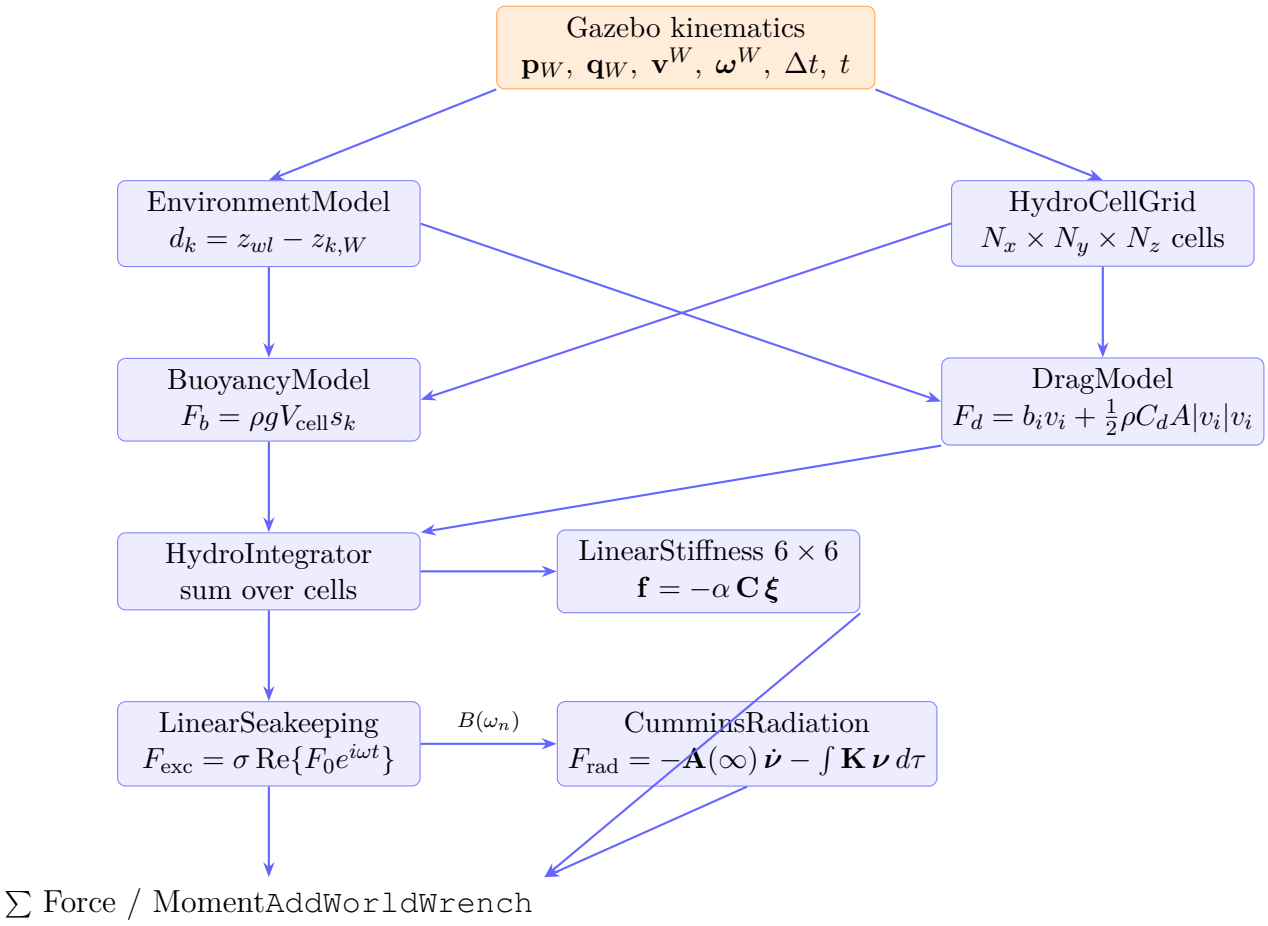

Figure 1: System data flow: from Gazebo kinematics to force composition

2 EnvironmentModel – Environment Model

2.1 Purpose

`EnvironmentModel` provides the fluid environment parameters to all other models. It is a static data layer – loaded from configuration and not recomputed each step.

2.2 Parameters

Table 2: `EnvironmentModel` parameters

Parameter	Default	Description
`water_level`	0.0 m	Calm water surface height (world Z)
ρ `fluid_density`	1000.0 kg/m^3	Fluid density
g `gravity`	9.81 m/s^2	Gravitational acceleration
`current_velocity`	$(0, 0, 0)$ m/s	Current velocity in world frame

2.3 Formulas

Cell centre depth:

$$d_k = z_{wl} - z_{k,W} \tag{4}$$

$d_k > 0$: cell is submerged. $d_k < 0$: cell is above water.

Relative velocity (current effect):

$$\mathbf{v}_{rel}^W = \mathbf{v}_{\text{body}}^W - \mathbf{v}_{\text{current}}^W \tag{5}$$

3 HydroCellGrid + HydroIntegrator – Cell Grid System

3.1 Grid Construction

The axis-aligned bounding box (AABB) of the body is read from Gazebo and divided into $N_x \times N_y \times N_z$ cells:

$$V_{\text{cell}} = \frac{V_{\text{AABB}}}{N_x N_y N_z} \tag{6}$$

$$h_{\text{cell}} \approx \frac{H_{\text{AABB}}}{N_z} \tag{7}$$

Cell position in world frame:

$$\mathbf{p}_k^W = \mathbf{p}_W + \mathbf{R}\,\mathbf{r}_k^{\text{body}} \tag{8}$$

3.2 Cell Kinematics

Rigid-body cell velocity:

$$\mathbf{v}_k^W = \mathbf{v}^W + \boldsymbol{\omega}^W \times (\mathbf{R}\,\mathbf{r}_k^{\text{body}}) \tag{9}$$

Body-frame relative velocity:

$$\mathbf{v}_{rel,k}^{\text{body}} = \mathbf{R}^T\Big(\mathbf{v}_k^W - \mathbf{v}_{\text{current}}^W\Big) \tag{10}$$

For each cell, `BuoyancyModel` and `DragModel` are invoked; forces are rotated to world frame and applied via `link.AddWorldForce(offset)`.

4 BuoyancyModel – Buoyancy Model

4.1 Purpose

Computes the submerged volume fraction and resulting buoyancy force for each cell, based on Archimedes' principle.

4.2 Inputs

Table 3: `BuoyancyModel` inputs

Input	Source	Description
d_k	EnvironmentModel	Cell centre depth (m)
h_{cell}	HydroCellGrid	Cell height (m)
V_{cell}	HydroCellGrid	Cell volume (m^3)
ρ	HydroConfig	Fluid density (kg/m^3)
g	HydroConfig	Gravity (m/s^2)

4.3 Formulas

Step 1 – Submergence fraction:

$$\boxed{s_k = \text{clamp}\left(\frac{d_k}{h_{\text{cell}}} + 0.5,\ 0,\ 1\right)} \tag{11}$$

- $s_k = 0$: cell fully above water
- $s_k = 1$: cell fully submerged
- $0 < s_k < 1$: free surface passes through cell

Step 2 – Submerged volume:

$$V_{\text{sub},k} = V_{\text{cell}} \cdot s_k \tag{12}$$

Step 3 – Buoyancy force (Archimedes):

$$\boxed{F^z_{b,k} = \rho\, g\, V_{\text{cell}}\, s_k} \tag{13}$$

Force direction is $+z$ only: $\mathbf{F}^W_{b,k} = (0,\, 0,\, F^z_{b,k})$.
Total buoyancy force and moment:

$$\mathbf{F}^W_{\text{buoy}} = \sum_{k:\, s_k > 0} \mathbf{F}^W_{b,k} \tag{14}$$

$$\boldsymbol{\tau}^W_{\text{buoy}} = \sum_{k:\, s_k > 0} \mathbf{r}^W_k \times \mathbf{F}^W_{b,k} \tag{15}$$

Equilibrium condition:

$$\rho\, g\, V_{\text{disp}} = m\, g \quad \Rightarrow \quad V_{\text{disp}} = \frac{m}{\rho} \tag{16}$$

4.4 Numerical Values (Reference Vessel)

Table 4: Buoyancy model parameters for the reference USV

Quantity	**Formula**	**Value**
Hull dimensions	$L \times B \times D$	$2.2 \times 1.1 \times 0.5$ m
Grid	$N_x \times N_y \times N_z$	$14 \times 8 \times 4 = 448$ cells
Cell volume	V/N	5.45×10^{-4} m^3
Cell height	H/N_z	0.125 m
Waterplane area	$L \times B$	2.42 m^2
Static draft	$m/(\rho LB)$	≈ 0.145 m

5 DragModel – Viscous Resistance Model

5.1 Purpose

Computes viscous resistance from the fluid for each submerged cell, including both linear and quadratic (Morison) terms.

5.2 Inputs

Table 5: DragModel inputs

Input	Source	Description
$\mathbf{v}_{rel,k}^{\text{body}}$	HydroIntegrator	Body-frame relative velocity (m/s)
$A_{\text{cell},i}$	HydroCellGrid	Cell cross-section area in direction i (m^2)
b_i	HydroConfig	Linear damping coefficient [N·s/m]
$C_{d,i}$	HydroConfig	Quadratic drag coefficient [–]
s_k	BuoyancyModel	Submergence fraction $[0, 1]$
ρ	HydroConfig	Fluid density (kg/m^3)

5.3 Formulas

For $i \in \{\text{surge, sway, heave}\}$:

$$\boxed{F_{d,i,k}^{\text{body}} = \underbrace{-b_i\, v_{rel,i,k}}_{\text{linear}} - \underbrace{\tfrac{1}{2}\rho\, C_{d,i}\, A_{\text{cell},i}\, |v_{rel,i,k}|\, v_{rel,i,k}}_{\text{quadratic (Morison)}}} \tag{17}$$

Scaling for partially submerged cells:

$$F_{d,i,k}^{\text{eff}} = F_{d,i,k}^{\text{body}} \cdot s_k \tag{18}$$

Rotation to world frame:

$$\mathbf{F}_{d,k}^{W} = \mathbf{R}\, \mathbf{F}_{d,k}^{\text{eff}} \tag{19}$$

5.4 Profile Parameters

Table 6: Damping parameters across configuration profiles

Profile	b_x [N·s/m]	b_y	b_z
baseline	18	55	80
low_damping	6	18	25
high_damping	30	90	130

$C_{d,x} = 0.9$, $C_{d,y} = 1.3$, $C_{d,z} = 2.0$ across all profiles.

6 LinearStiffness6x6 – Hydrostatic Stiffness Matrix

6.1 Purpose

Computes restoring forces and moments opposing the vessel's deviation from its equilibrium position, expressed as a full 6×6 stiffness matrix.

6.2 6x6 Stiffness Matrix – Finite Differences

At the first simulation step, the reference pose $(\mathbf{p}_{\text{ref}}, \mathbf{q}_{\text{ref}})$ is recorded. For each DOF $j \in \{0, \ldots, 5\}$ a $\pm$ perturbation is applied and buoyancy forces are rotated to body frame:

$$\mathbf{F}^{\text{body}}(\pm\Delta\xi_j) = \mathbf{R}_{\text{ref}}^T \, \mathbf{F}^{\text{world}}(\pm\Delta\xi_j) \tag{20}$$

Central finite difference stiffness coefficient:

$$\boxed{C_{ij} = -\frac{F_i^{\text{body}}(+\Delta\xi_j) - F_i^{\text{body}}(-\Delta\xi_j)}{2\,\Delta\xi_j}} \tag{21}$$

Table 7: Per-column perturbation vectors (Δs: translational step, Δa: angular step)

Column j	**DOF**	**Step size**
0	surge	$\Delta s = 0.01$ m
1	sway	Δs
2	heave	Δs
3	roll	$\Delta a = 10^{-3}$ rad
4	pitch	Δa
5	yaw	Δa

6.3 Restoring Wrench Computation

Instantaneous 6-DOF displacement relative to reference pose:

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{R}_{\text{ref}}^T(\mathbf{p}^W - \mathbf{p}_{\text{ref}}) \\ \phi_e,\ \theta_e,\ \psi_e \end{bmatrix} \in \mathbb{R}^6 \tag{22}$$

$$\boxed{f_i^{\text{body}} = -\alpha \sum_{j=0}^{5} C_{ij}\,\xi_j, \quad i = 0, \ldots, 5} \tag{23}$$

World-frame transformation:

$$\mathbf{F}_{\text{rest}}^W = \mathbf{R}_{\text{ref}}\,\mathbf{f}_{0:2}^{\text{body}}, \qquad \boldsymbol{\tau}_{\text{rest}}^W = \mathbf{R}_{\text{ref}}\,\mathbf{f}_{3:5}^{\text{body}} \tag{24}$$

6.4 Analytical Comparison

$$C_{33} = \rho\, g\, A_{wp} \qquad \text{(waterplane stiffness)} \tag{25}$$

$$C_{44} = \rho\, g\, V\, \overline{GM}_T \qquad \text{(roll restoring)} \tag{26}$$

$$C_{55} = \rho\, g\, V\, \overline{GM}_L \qquad \text{(pitch restoring)} \tag{27}$$

Table 8: Stiffness matrix validation results

Term	Analytical	Numerical	Error
C_{33}	23 738 N/m	23 740 N/m	$< 0.01\%$
C_{44}	1 771 Nm/rad	1 771 Nm/rad	$< 0.01\%$
C_{55}	8 941 Nm/rad	8 941 Nm/rad	$< 0.01\%$
C_{11}, C_{22}	≈ 0	≈ 0	–
$C_{35} \approx C_{53}$	reciprocity	< 150 Nm/m	–

7 LinearSeakeepingModel – Frequency-Domain Seakeeping

7.1 Governing Equation

$$\boxed{\mathbf{F}_{\text{sk}} = \underbrace{-\mathbf{A}(\omega)\,\dot{\boldsymbol{\nu}}}_{\text{added mass}} - \underbrace{\mathbf{B}(\omega)\,\boldsymbol{\nu}}_{\text{radiation damping}} + \underbrace{\text{Re}\Big\{\mathbf{F}_{\text{exc}}(\omega)\,e^{i\omega t}\Big\}}_{\text{wave excitation}}} \tag{28}$$

Excitation term in explicit form:

$$F_{\text{exc},i}(t) = \sigma\Big[F^{\text{Re}}_{\text{exc},i}\cos(\omega t) - F^{\text{Im}}_{\text{exc},i}\sin(\omega t)\Big] \tag{29}$$

7.2 Inputs

Table 9: Seakeeping model inputs

Input	Source	Description
YAML coefficient file	Disk	$\mathbf{A}(\omega_n)$, $\mathbf{B}(\omega_n)$, $\mathbf{F}_{\text{exc}}(\omega_n)$
ω	scalar	Evaluation frequency (rad/s)
$\boldsymbol{\nu}$	HydroCellSystem	Body-frame 6-DOF velocity
$\dot{\boldsymbol{\nu}}$	HydroCellSystem	Body-frame 6-DOF acceleration
t	Gazebo	Simulation time (s)
σ	scalar	Wave amplitude scaling factor

7.3 YAML Coefficient Format

```
schema_version: 1
# DOF order: [surge, sway, heave, roll, pitch, yaw]
frequencies:
  - omega: 0.5              # rad/s
```

```
    added_mass:             # 6x6 matrix [kg]
      - [120.0, 0.0, 0.0, 0.0, 0.0, 0.0]
      ...
    damping:                # 6x6 [N*s/m]
      - [45.0, ...]
      ...
    excitation_re: [Fx, Fy, Fz, Mx, My, Mz]   # [N] or [N*m]
    excitation_im: [...]
  - omega: 1.0
    ...
```

Compatible with AQWA, Nemoh, or WAMIT outputs after conversion.

7.4 Linear Interpolation

$$\alpha = \frac{\omega - \omega_{lo}}{\omega_{hi} - \omega_{lo}}, \qquad \mathbf{A}(\omega) = (1-\alpha)\,\mathbf{A}(\omega_{lo}) + \alpha\,\mathbf{A}(\omega_{hi}) \tag{30}$$

7.5 Operating Mode Selection

- **Legacy mode** (`cummins_radiation: false`): Full Equation (**??**) applied.
- **Cummins mode** (`cummins_radiation: true`): Only the excitation term is computed; the radiation part is replaced by `CumminsRadiationModel`.

8 CumminsRadiationModel – Time-Domain Radiation Model

8.1 Cummins Equation

$$\boxed{\Big[\mathbf{M} + \mathbf{A}(\infty)\Big]\,\dot{\boldsymbol{\nu}}(t) + \int_0^t \mathbf{K}(t-\tau)\,\boldsymbol{\nu}(\tau)\,d\tau + \mathbf{C}\,\boldsymbol{\xi}(t) = \mathbf{F}_{\text{exc}}(t) + \mathbf{F}_{\text{vis}}(t)} \tag{31}$$

Table 10: Cummins equation terms

Term	Dimension	Description
$\mathbf{M}$	$\mathbb{R}^{6\times 6}$	Body mass + inertia matrix
$\mathbf{A}(\infty)$	$\mathbb{R}^{6\times 6}$	Added mass at $\omega \to \infty$
$\mathbf{K}(t)$	$\mathbb{R}^{6\times 6}$	Retardation (memory) kernel
$\mathbf{C}$	$\mathbb{R}^{6\times 6}$	Hydrostatic stiffness matrix
$\mathbf{F}_{\text{exc}}(t)$	$\mathbb{R}^{6}$	First-order wave excitation force
$\mathbf{F}_{\text{vis}}(t)$	$\mathbb{R}^{6}$	Viscous forces

Since Gazebo's rigid-body dynamics already computes the $\mathbf{M}\dot{\boldsymbol{\nu}}$ term, the plugin applies *only* the $(-\mathbf{A}(\infty)\dot{\boldsymbol{\nu}} - I(t))$ contribution.

8.2 Retardation Kernel K(t) – Computation

Fourier cosine transform of $B(\omega)$ samples (Ogilvie, 1964):

$$\boxed{K_{ij}(t) = \frac{2}{\pi}\int_0^\infty B_{ij}(\omega)\cos(\omega t)\,d\omega} \tag{32}$$

Numerical approximation – non-uniform trapezoidal rule:

$$K_{ij}(t_k) \approx \frac{2}{\pi}\sum_{n=0}^{N_\omega-2}\frac{\omega_{n+1}-\omega_n}{2}\left[B_{ij}(\omega_n)\cos(\omega_n t_k) + B_{ij}(\omega_{n+1})\cos(\omega_{n+1}t_k)\right] \tag{33}$$

Kernel tail truncation:

$$|K_{ij}(t_k)| < 10^{-6}\max_{ij}|K_{ij}(0)| \implies \text{kernel truncated at this point} \tag{34}$$

8.3 A(infinity) – Ogilvie Relation

$$\boxed{A_{ij}(\infty) = A_{ij}(\omega_{\max}) + \frac{1}{\omega_{\max}}\int_0^T K_{ij}(t)\sin(\omega_{\max}\,t)\,dt} \tag{35}$$

8.4 Convolution Integration – Each Step

Each step $\boldsymbol{\nu}(t)$ is appended to a circular buffer (`std::deque`). Maximum history length: $N_{\max} = \lceil T_{\max}/\Delta t_k\rceil + 1$.

$$I_i(t) \approx \sum_{k=0}^{N-1} w_k\,K_{ij}(t_k)\,\nu_j(t-t_k)\,\Delta t_k \tag{36}$$

Radiation force applied by the plugin:

$$\boxed{F_{\text{rad},i} = -\sum_{j=0}^{5} A_{ij}(\infty)\,\dot{\nu}_j(t) \;-\; I_i(t)} \tag{37}$$

8.5 Equivalence with Frequency-Domain Model

For single-frequency sinusoidal motion, the Cummins equation reduces to the frequency-domain model:

$$\int_{-\infty}^{t} K(t-\tau)\,e^{i\omega\tau}\,d\tau = \Big[\mathbf{B}(\omega) - i\omega\Big(\mathbf{A}(\infty) - \mathbf{A}(\omega)\Big)\Big]\,e^{i\omega t} \tag{38}$$

Therefore the Cummins model is the **general form** of the frequency-domain model.

9 Inter-Model Dependencies and Data Flow

9.1 Hierarchical Dependency Graph

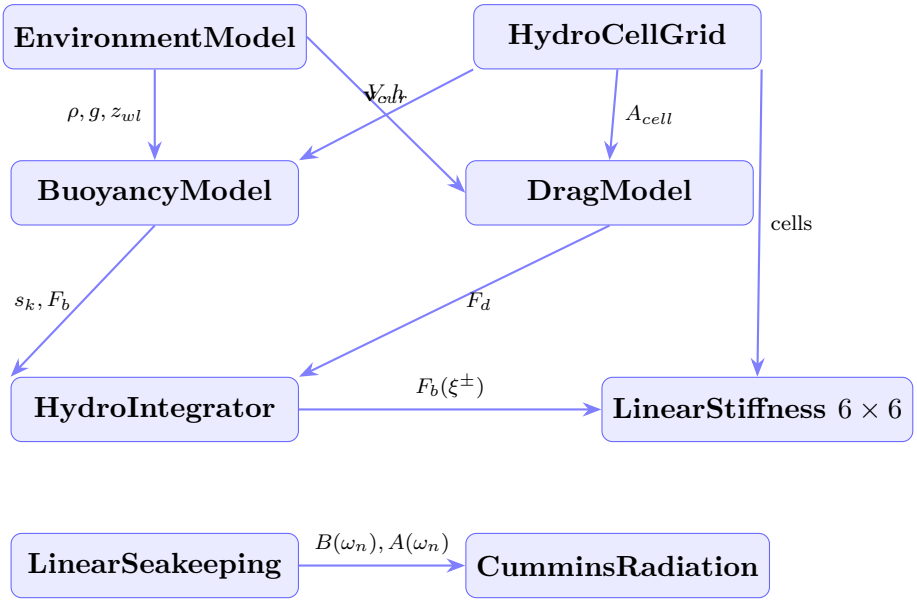

Figure 2: Inter-model data dependency graph

9.2 Computation Order (Each Step)

1. **Kinematics derivation:** Gazebo data → body-frame $\boldsymbol{\nu}$, $\dot{\boldsymbol{\nu}}$
2. **Cell loop:** For each k: $d_k \to s_k \to F_{b,k}$ and $F_{d,k} \to$ `AddWorldForce`
3. **Hydrostatic stiffness:** $\boldsymbol{\xi} \to \mathbf{C}\,\boldsymbol{\xi} \to$ `AddWorldWrench`
4. **Seakeeping / Cummins selection:**
   - *Cummins active:* $\boldsymbol{\nu} \to I(t)$, $\dot{\boldsymbol{\nu}} \to -\mathbf{A}(\infty)\dot{\boldsymbol{\nu}}$, excitation separate → two wrenches
   - *Legacy mode:* $-\mathbf{A}\dot{\boldsymbol{\nu}} - \mathbf{B}\boldsymbol{\nu} + F_{\text{exc}} \to$ single wrench

9.3 Force Budget

Table 11: Total force components

Component	Model	Profile	Physical Origin
$\mathbf{F}_{\text{buoy}}$	BuoyancyModel	all	Archimedes buoyancy
$\mathbf{F}_{\text{drag}}$	DragModel	all	Viscous + form resistance
$\mathbf{F}_{\text{stiff}}$	LinearStiffness	stiffness_on	Hydrostatic restoring
$\mathbf{F}_{\text{rad}}$	CumminsRadiation	cummins_on	Radiation inertia + memory
$\mathbf{F}_{\text{exc}}$	LinearSeakeeping	seakeeping_on	Wave diffraction (1st order)

10 Coefficient Estimation Module – `coeff_estimator`

10.1 Overview and Purpose

`coeff_estimator` is a Python utility package that automatically computes all hydrodynamic coefficients required by the `usv_hydro` plugin from a given hull geometry. It is located at `usv_sim_ws/src/usv_hydro/tools/coeff_estimator/` and can be run independently.

Input: Hull dimensions (L, B, D, m, I_{xx}, I_{yy}, I_{zz}), or an SDF / STL file.

Output: A `hydro_profiles.yaml` profile entry + `seakeeping_coeffs_<name>.yaml`.

10.2 Module Architecture

Table 12: `coeff_estimator` module files

File	Responsibility
`geometry_loader.py`	Converts SDF / STL / box parameters to a `ShipGeometry` object; derives T, C_B, A_{wp}, S_{wet}
`viscous_drag_estimator.py`	Computes C_d via Hoerner (1965), b_i via ITTC-1957, A_{ii} via Lewis section method
`bem_solver.py`	Produces $A(\omega)$, $B(\omega)$, $F_{\text{exc}}(\omega)$ via Capytaine BEM or closed-form Froude-Krylov
`yaml_writer.py`	Writes results as a profile YAML entry and seakeeping coefficient YAML
`estimate_coeffs.py`	CLI entry point (`box` / `sdf` / `stl` subcommands)

10.3 GeometryLoader – Geometry Loader

The `ShipGeometry` dataclass automatically derives the following:

$$T = \frac{m}{\rho\, L\, B} \qquad \text{(static equilibrium draft)} \tag{39}$$

$$C_B = \frac{m/\rho}{L\, B\, T} \qquad \text{(block coefficient)} \tag{40}$$

$$A_{wp} = L\, B \qquad \text{(waterplane area)} \tag{41}$$

$$S_{wet} \approx 1.025\sqrt{V_{\text{disp}}\, L} \qquad \text{(Denny-Mumford wetted surface)} \tag{42}$$

Automatic grid suggestion:

$$N_x = \text{clip}(\text{round}(6L),\, 4,\, 20), \quad N_y = \text{clip}(\text{round}(6B),\, 4,\, 16), \quad N_z = 4 \tag{43}$$

10.4 ViscousDragEstimator – Empirical Resistance

10.4.1 C_d – Hoerner Form Drag

Hoerner (1965) form drag table with C_B correction:

$$C_{d,x} = 0.10 + 1.2\,(C_B - 0.50)^2 \qquad \text{(surge – bow section)} \tag{44}$$

$$C_{d,y} = 1.10 + 0.5\,(C_B - 0.60) \qquad \text{(sway – side surface)} \tag{45}$$

$$C_{d,z} = 1.10 + \frac{0.8}{L/B} \qquad \text{(heave – bottom surface)} \tag{46}$$

10.4.2 b_x – Surge Linear Damping (ITTC-1957)

Reynolds number and ITTC-1957 friction coefficient:

$$Re = \frac{V_{des}\, L}{\nu}, \qquad \nu = 1.14 \times 10^{-6}\ \text{m}^2/\text{s} \tag{47}$$

$$C_f = \frac{0.075}{(\log_{10} Re - 2)^2} \tag{48}$$

Frictional resistance and linear coefficient:

$$R_f = \tfrac{1}{2}\rho\, V_{des}^2\, S_{wet}\, C_f\, (1 + k_1) \tag{49}$$

$$b_x = \max\left(\frac{R_f}{V_{des}},\ \frac{m + A_{11}}{\tau_{\text{surge}}}\right) \tag{50}$$

10.4.3 b_y, b_z – Sway and Heave Damping

From the heave critical damping ratio:

$$C_{33} = \rho\, g\, A_{wp} \tag{51}$$

$$\omega_{n,3} = \sqrt{\frac{C_{33}}{m + A_{33}}} \tag{52}$$

$$b_z = 2\,\zeta_{\text{heave}}\,\omega_{n,3}\,(m + A_{33}) \tag{53}$$

Sway with characteristic decay time:

$$b_y = \frac{m + A_{22}}{\tau_{\text{sway}}}, \qquad \tau_{\text{sway}} = \max\Big(5.0,\ \frac{B}{2V_{des}}\Big) \tag{54}$$

10.4.4 A_{ii} – Added Mass (Lewis Section)

$$A_{11} = \max(0.02,\ C_B - 0.45)\, m \qquad \text{(surge)} \tag{55}$$

$$A_{22} = \tfrac{\pi}{2}\rho\,(B/2)^2\, L \times 0.80 \qquad \text{(sway)} \tag{56}$$

$$A_{33} = \tfrac{\pi}{4}\rho\,(B/2)^2\, L \times 0.70 \qquad \text{(heave)} \tag{57}$$

$$A_{44} = 0.30\, I_{xx} \qquad \text{(roll)} \tag{58}$$

10.5 BEMSolver – Wave Force Coefficients

10.5.1 Capytaine Mode (full BEM)

When Capytaine is installed, a panel BEM solution is performed: the hull mesh (box or STL) is generated, radiation and diffraction problems are solved for each ω, and $A(\omega)$, $B(\omega)$, $F_{\text{exc}}(\omega)$ are assembled as (6×6) matrices.

10.5.2 Empirical Mode (no Capytaine)

Closed-form frequency-dependent models:

$$A_{ii}(\omega) = A_{ii}^0 \cdot \frac{1 - \tanh\Big((\omega - \omega_c)/\Delta\omega\Big)}{2} \tag{59}$$

$$B_{ii}(\omega) = B_{\text{peak}} \cdot \exp\left(-\frac{(\omega - \omega_c)^2}{2\,\Delta\omega^2}\right) \tag{60}$$

Froude-Krylov excitation forces:

$$F_{\text{exc},3} = \rho g A_{wp} \qquad \text{(heave, constant)} \tag{61}$$

$$F_{\text{exc},1} = i\,\rho g k\, V_{\text{disp}} \qquad \text{(surge, inertia component)} \tag{62}$$

$$F_{\text{exc},5} = -F_{\text{exc},3}\,\frac{L\,k}{12} \qquad \text{(pitch)} \tag{63}$$

where $k = \omega^2/g$ is the deep-water wave number.

10.6 CLI Usage Examples

```
# 1. From box parameters (usv_minimal.sdf values)
python3 estimate_coeffs.py box \
    --L 2.2 --B 1.1 --D 0.5 --mass 350 \
    --ixx 42.58 --iyy 148.46 --izz 176.46 \
    --name usv_minimal --out output/ --bem

# 2. Automatic read from SDF
python3 estimate_coeffs.py sdf \
    --sdf ../usv_description/sdf/usv_minimal.sdf \
    --name usv_minimal --out output/

# 3. STL mesh + full BEM (Capytaine required)
python3 estimate_coeffs.py stl \
    --stl hull.stl --mass 350 --bem \
    --omega-min 0.3 --omega-max 3.14 --n-omega 12 \
    --name my_hull --out output/
```

10.7 Reference Vessel Results

The table below compares estimation results obtained with `usv_minimal.sdf` ($2.2 \times 1.1 \times 0.5$ m, 350 kg) against the `baseline` profile values in `hydro_profiles.yaml`.

Table 13: Estimated vs. baseline profile values

Quantity	**Estimated**	**baseline**	**Note**
$C_{d,x}$ (surge)	0.343	0.90	Low for C_B=1; expected
$C_{d,y}$ (sway)	1.275	1.30	Consistent
$C_{d,z}$ (heave)	1.500	2.00	Conservative estimate
b_x [N·s/m]	18.1	18.0	Nearly identical
b_y [N·s/m]	237	55	Large Lewis A_{22} contribution
b_z [N·s/m]	1237	80	ζ=0.15 + high C_{33}
Static draft T	0.1446 m	0.145 m	$< 0.05\%$ error
A_{wp}	2.42 m^2	2.42 m^2	Exact
$F_{exc,3}$ (heave)	23 740 N/m	23 738 (analytical)	$< 0.01\%$

Table 14: Empirical added mass estimates

DOF	Value	Unit
A_{11} surge	192.5	kg
A_{22} sway	836.3	kg
A_{33} heave	365.9	kg
A_{44} roll	12.77	kg·m^2
A_{55} pitch	129.83	kg·m^2
A_{66} yaw	44.12	kg·m^2

11 Configuration Guide

```
defaults:
  fluid_density: 1000.0     # [kg/m^3]
  gravity: 9.81             # [m/s^2]
  water_level: 0.0          # [m]
  link_name: "hull_link"
  cells: [14, 8, 4]
  cd: [0.9, 1.3, 2.0]
  drag:
    linear_total: [18.0, 55.0, 80.0]
    scale_by_cell_count: true

profiles:
  stiffness_on:
    hydrostatic_stiffness:
      enabled: true
      scale: 1.0
      heave_step: 0.01       # [m]
      angle_step: 0.001      # [rad]

  cummins_on:
    linear_seakeeping:
      enabled: true
      coeffs_file: "seakeeping_coeffs_example.yaml"
      excitation_omega: 0.8  # [rad/s]
      excitation_scale: 1.0
    cummins_radiation:
      enabled: true
      kernel_max_t: 20.0     # [s]
      kernel_dt: 0.0         # 0 = use sim step
```

12 Testing and Validation

Table 15: Unit test coverage (21/21 passed, 3.04 s)

Test Suite	Count	Validated
`BuoyancyModelTest`	3	Archimedes, partial submergence, equilibrium draft
`DragModelTest`	3	Current damping, linear/quadratic components
`HydroScenarioTest`	6	Equilibrium, roll/pitch moment, $C_{11}/C_{22} \approx 0$, $C_{35} \approx C_{53}$
`CumminsRadiationModelTest`	6	Kernel construction, zero-velocity test, $A(\infty)$, memory accumulation
`HydroConfigTest`	3	YAML validation

13 Current Limitations and Development Roadmap

13.1 Current Limitations

- AABB-based box hull approximation; no true cross-sectional geometry
- Single harmonic wave; irregular sea states (JONSWAP) not supported
- No Munk moment, cross-flow lift
- No propulsion or rudder model

13.2 Phase 2 – MMG Manoeuvring Model

$$N_{\text{Munk}} = -\tfrac{1}{2}\rho(m_{11} - m_{22})uv \tag{64}$$

$$Y_{cf} = -\tfrac{1}{2}\rho C_{dL}\, L\, d\, |v|v \tag{65}$$

$$T = \rho n^2 D^4 K_t(J), \quad J = \frac{u(1-w)}{nD} \tag{66}$$

A Symbol List

Symbol	Unit	Definition
ρ	kg/m^3	Fluid density
g	m/s^2	Gravitational acceleration
$\mathbf{R}$	–	Rotation matrix
$\boldsymbol{\nu} \in \mathbb{R}^6$	m/s, rad/s	Body-frame 6-DOF velocity
$\dot{\boldsymbol{\nu}} \in \mathbb{R}^6$	m/s^2	Body-frame 6-DOF acceleration
$\boldsymbol{\xi} \in \mathbb{R}^6$	m, rad	6-DOF displacement
s_k	–	Submergence fraction $[0, 1]$
V_{cell}	m^3	Cell volume
d_k	m	Cell depth
b_i	N·s/m	Linear damping
$C_{d,i}$	–	Morison drag coefficient
$\mathbf{C} \in \mathbb{R}^{6\times 6}$	N/m	Hydrostatic stiffness
$\mathbf{A}(\omega)$	kg	Added mass matrix
$\mathbf{B}(\omega)$	N·s/m	Radiation damping matrix
$\mathbf{A}(\infty)$	kg	Added mass at infinite frequency
$\mathbf{K}(t)$	N·s/m^2	Retardation kernel
$T_{\max}$	s	Cummins memory window
Δt_k	s	Kernel time step
C_B	–	Block coefficient
A_{wp}	m^2	Waterplane area
S_{wet}	m^2	Wetted surface area
T	m	Static draft
k	1/m	Deep-water wave number ω^2/g

B Reference Vessel Parameters

Parameter	Value	Unit
$L \times B \times D$	$2.2 \times 1.1 \times 0.5$	m
Mass m	350	kg
I_{xx}	42.58	kg·m^2
I_{yy}	148.46	kg·m^2
I_{zz}	176.46	kg·m^2
ρ	1000	kg/m^3
Static draft	0.145	m
A_{wp}	2.42	m^2